

02443 Stochastic Simulation

Day 2: Sampling from continuous distributions

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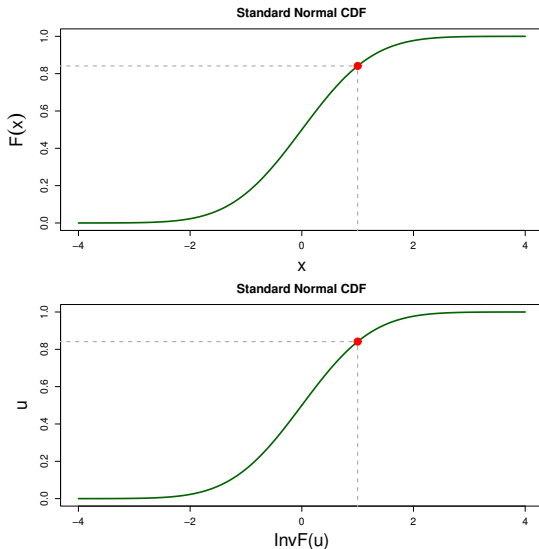
Agenda

- 1 Inverse transformation techniques
- 2 Mathematical methods
- 3 Composition methods
- 4 Rejection methods
- 5 Exercise 3

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The inverse transformation



The inverse transformation method

Consider the random variable $U = F_X(X)$. Then if F_X is invertible

$$F_U(u) = \mathbb{P}(U \leq u) = \mathbb{P}(F_X(X) \leq u) = \mathbb{P}(X \leq F_X^{-1}(u)) = F_X(F_X^{-1}(u)) = u.$$

Consequently, $U \sim \text{U}(0, 1)$.

Comment on p -values

Since $1 - U \sim \text{U}(0, 1)$, we have that $1 - F_X(X)$ is also uniformly distributed on the unit interval. But $1 - F_X(X)$ is exactly a one-sided random p -value.

Conversely, for $Y = F_X^{-1}(U)$ we have that

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(F_X^{-1}(U) \leq y) = \mathbb{P}(U \leq F_X(y)) = F_U(F_X(y)) = F_X(y).$$

Consequently, Y has the same distribution as X .

Example: The exponential distribution

The exponential distribution is foundational to stochastic models and probability theory. It is e.g. the distribution of the interarrival times in a Poisson process.

Recap

For $X \sim \exp(\lambda)$, the distribution function is given by $F_X(x) = 1 - e^{-\lambda x}$.

We can derive that

$$F_X^{-1}(u) = -\frac{1}{\lambda} \log(1 - u)$$

leading to

$$Y = F_X^{-1}(U) = -\frac{\log(U)}{\lambda} \sim \exp(\lambda).$$

Example: The Pareto distribution

The Pareto distribution is often used in social sciences e.g., to describe income and wealth distributions in society and communities.

Recap

For $X \sim \text{Pareto}(k, \beta)$, the distribution function is given by $F_X(x) = 1 - (\beta/x)^k$ for $x \geq \beta$. In addition,

$$\mathbb{E}[X] = \frac{k}{k-1}\beta, \quad k > 1 \quad \text{and} \quad \mathbb{V}[X] = \frac{k}{(k-1)^2(k-2)}\beta^2, \quad k > 2.$$

We can derive that

$$F_X^{-1}(u) = \beta(1-u)^{-\frac{1}{k}}$$

leading to

$$Y = F_X^{-1}(U) = \beta U^{-\frac{1}{k}} \sim \text{Pareto}(k, \beta).$$

For a Pareto distribution with support on the non-negative reals:

$$F_X(x) = 1 - \left(1 + \frac{x}{\beta}\right)^{-k} \implies Y = F_X^{-1}(U) = \beta(U^{-\frac{1}{k}} - 1).$$

Example: Gaussian distribution

The most widely applied distribution in statistics and probability theory.

For $X \sim N(\mu, \sigma^2)$, we can decompose X as $X = \mu + \sigma Z$, where Z has a standard normal distribution with cdf Φ .

Therefore, we can generate random variates by $Z = \Phi^{-1}(U)$ and obtain samples of X through the above relationship.

Example: Rayleigh distribution

Consider $X_1, X_2 \sim N(0, 1)$ and define $R = \sqrt{X_1^2 + X_2^2}$. Then the distribution function is given by

$$F_R(r) = \mathbb{P}(R \leq r) = 1 - e^{-\frac{r^2}{2}}.$$

Furthermore $S = R^2 = X_1^2 + X_2^2 \sim \chi^2(2)$, which means that $S \sim \exp(1/2)$. Both R and S can be generated easily by the inversion method. This enables another method for generating random variables from a normal distribution - even two independent random variables:

$$X_1 = R \cos(2\pi\theta) \quad \text{and} \quad X_2 = R \sin(2\pi\theta)$$

with $\theta \sim U(0, 1)$ independent of R .

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Mathematical methods and the Box-Muller method

The result on the last slide is known as the Box-Muller method. In general, we can apply transformation and other techniques to obtain random variables from certain distributions.

Box-Muller

The Box-Muller is an example of a transformation from polar coordinates into Cartesian coordinates that can be used to sample independent Gaussian random variables.

$$\begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} = \sqrt{-2\log(U_1)} \begin{bmatrix} \cos(2\pi U_2) \\ \sin(2\pi U_2) \end{bmatrix}$$

Central Limit Theorem

Similarly, the Central Limit Theorem can be used to generate Gaussian random variables.

$$X = \frac{\sum_{i=1}^n U_i - \frac{n}{2}}{\sqrt{\frac{n}{12}}}.$$

The log-normal and multivariate Gaussian

The access to random variables following Gaussian distributions allows us to generate new random variables.

The log-normal

Let $Z \sim N(0, 1)$. Then construct $X = \alpha + \beta Z$ and define $Y = e^X$. Then

$$\log(Y) \sim N(\alpha, \beta^2) \quad \text{and} \quad Y \sim LN(\alpha, \beta^2).$$

The multivariate normal

We can generate a random vector with a multivariate normal distribution: $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ by letting

$$X_i = \mu_i + \sum_{j=1}^i c_{ij} Z_j,$$

where the coefficients are taken from the Cholesky factorisation of $\boldsymbol{\Sigma} = \mathbf{C}\mathbf{C}^T$.

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Composition method: Hyperexponential distribution

Consider the hyperexponential distribution with distribution function

$$F(x) = 1 - \sum_{i=1}^m p_i e^{-\lambda_i x} = \sum_{i=1}^m p_i \left(1 - e^{-\lambda_i x}\right).$$

The distribution is a mixture distribution of m independent exponential distributions.

Interpretation for simulation

Let I take values in $\{1, 2, \dots, m\}$ with distribution $P(I = i) = p_i$, and let $X_i \sim \exp(\lambda_i)$. Then

$$Z = X_I$$

has the desired hyperexponential distribution.

Generation procedure

- ① Generate I from the distribution $\{p_i\}$
- ② Generate Z :

$$Z = -\frac{1}{\lambda_I} \log(U).$$

Composition method: Erlang distribution

The Erlang distribution is a special case of the Gamma distribution with an integer-valued shape parameter.

Interpretation

An Erlang-distributed random variable can be represented as the sum of independent exponential random variables.

If $X_i \sim \exp(\lambda)$, $i = 1, \dots, n$, are independent, then

$$Y = \sum_{i=1}^n X_i$$

follows an Erlang(n)-distribution.

Hence, we can generate random variables from an Erlang(n)-distribution by the following procedure:

$$Y = \sum_{i=1}^n -\frac{1}{\lambda} \log(U_i) = -\frac{1}{\lambda} \log \left(\prod_{i=1}^n U_i \right).$$

Composition method: General form

Suppose

$$f_X(x) = \int f_{X|Y}(x | Y = y) f_Y(y) dy.$$

Interpretation

- Y has density $f_Y(y)$.
- Given $Y = y$, the random variable X has conditional density $f_{X|Y}(x | Y = y)$.

A common example is

$$X | Y = \mu \sim N(\mu, \sigma^2),$$

where Y acts as a random parameter.

Generation procedure

- 1 Generate Y from $f_Y(y)$.
- 2 Generate X from $f_{X|Y}(x | Y = y)$ using the generated value of Y .

Example

Consider $Y \sim \exp(\mu)$. The conditional distribution of X given $Y = y$ is an exponential distribution with rate y .

The marginal density of X can then be found as

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|Y=y)f_Y(y)dy = \int_0^{\infty} ye^{-yx}\mu e^{-\mu y}dy = \frac{\mu}{(\mu+x)^2}, \quad x > 0.$$

Then the distribution function becomes

$$F_X(x) = 1 - \left(1 + \frac{x}{\mu}\right)^{-1}$$

i.e., X follows a Pareto distribution with support on the non-negative reals and parameters $k = 1$ and $\beta = \mu$.

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Acceptance-rejection method

Suppose we wish to generate a random variable with density f , but it is much easier to generate a random variable with density g (on the same sample space). Assume that

$$\frac{f(y)}{g(y)} \leq c$$

for all y , where c is a finite constant.

Algorithm

- ① Generate Y with density g .
- ② Generate $U \sim U(0, 1)$.
- ③ If $U \leq \frac{f(Y)}{cg(Y)}$, accept and set $X = Y$.
- ④ Otherwise, reject the proposal and return to Step 1.

The acceptance probability is proportional to

$$\frac{f(y)}{cg(y)},$$

which ensures that the accepted observations follow the target density f .

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Exercise 3

Part 1

Generate simulated observations from the following distributions:

- 1 Exponential distribution.
- 2 Normal distribution (at least using the Box–Muller method).
- 3 Pareto distribution with $\beta = 1$.

For the Pareto distribution, experiment with different values of the shape parameter:

$$k = 2.05, \quad k = 2.5, \quad k = 3, \quad k = 4.$$

Verify the results by comparing histograms with the corresponding theoretical distributions and perform appropriate goodness-of-fit tests (i.e., tests for distribution type).

Exercise 3 (continued)

Part 2

Consider the Pareto distribution with support on $[\beta, \infty)$. Compare the simulated sample mean and variance with their theoretical values

$$\mathbb{E}[X] = \beta \frac{k}{k-1}, \quad k > 1,$$

and

$$\mathbb{V}[X] = \beta^2 \frac{k}{(k-1)^2(k-2)}, \quad k > 2.$$

Discuss any discrepancies that you observe and explain their possible causes.

Exercise 3 (continued)

Part 3

For the normal distribution, generate 100 confidence intervals for:

- the mean,
- the variance.

Each confidence interval should be based on a sample of size $n = 10$, and all confidence intervals should have confidence level 95%.

Present and discuss the results. In particular, compare the observed coverage with the nominal confidence level.

Part 4

Generate observations from the Pareto distribution using the composition method.

Compare this approach with the direct inversion method used previously.